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2027 CYCLE

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CUET UG

Biology

Microbes in Human Welfare

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VIII)	Skills the questions test
Microbes in household products	Microbes in household products	Matching each product (curd, batter, bread, cheese) to its microbe and to the one-line change the microbe makes.
Industrial products — beverages	Microbes in industrial products – **beverages**, antibiotics, enzymes and bioactive molecules	Naming the yeast that ferments malted cereals and fruit juices into ethanol.
Industrial products — antibiotics	Microbes in industrial products – beverages, **antibiotics**, enzymes and bioactive molecules	Telling the penicillin story in order, and naming the diseases antibiotics brought under control.
Industrial products — enzymes	Microbes in industrial products – beverages, antibiotics, **enzymes** and bioactive molecules	Matching each enzyme (lipases, pectinases, proteases, streptokinase) to its everyday job.
Industrial products — bioactive molecules	Microbes in industrial products – beverages, antibiotics, enzymes and **bioactive molecules**	Linking Cyclosporin A and statins to their maker microbes and their medical roles.
Microbes in sewage treatment	Microbes in sewage treatment and **production of biogas**	Telling primary from secondary treatment, reading BOD, and defining flocs and activated sludge.
Production of biogas	Microbes in sewage treatment and **production of biogas**	Naming methanogens, the biogas mixture, the plant's parts, and why cattle dung is the feed.
Microbes as biocontrol agents	Microbes as **biocontrol agents** and biofertilisers	Matching each living agent to its target pest and explaining the organic-farming logic.
Microbes as biofertilisers	Microbes as biocontrol agents and **biofertilisers**	Separating symbiotic from free-living nitrogen

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VIII)	Skills the questions test
		fixers, and stating the mycorrhiza and cyanobacteria jobs.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VIII also carries the friendlier face of microbes: the examples in this topic are name-pairs (microbe → product) that questions like to check.

Meet the Microbes

Leave a bowl of milk out on a warm night and by morning it has gone sour and thick. Nobody added anything — invisible workers turned up on their own and changed the milk while feeding on it. That everyday surprise is this whole chapter: the same tiny workers that spoil food also make curd, bread, medicine and cooking gas for us. Before we meet them at work, here is who "they" are.

- **Microbe defined:** a microbe (microorganism) is any living thing too small to see without a microscope — you meet millions of them on your hands right now.
- **Four real kingdoms in one word:** bacteria, fungi, protozoa, and microscopic viruses (plus viroids and prions, which sit on the edge of life).
- **They live everywhere:** soil, water, air, inside your gut, under your fingernails — and even in hot springs at about 100 °C and in salty brines.
- **Most help, a few harm:** for every microbe that causes a disease, a huge number quietly recycle nutrients, ferment food and clean our sewage.
- **Fermentation is the magic word:** the process where microbes turn one substance into another while digesting it — milk to curd, juice to wine.

The microbe family you will meet

Group	What it is	Everyday example
Bacteria	Single-celled organisms with no membrane around their genetic material; they divide fast and live in almost every habitat.	**Lactobacillus** setting curd; bacteria cleaning sewage.

Group	What it is	Everyday example
Fungi	Yeasts are single cells; moulds grow as fuzzy threads. They absorb food from their surroundings.	Baker's yeast raising dough; the mould on old bread.
Protozoa	Single cells that swim and hunt smaller microbes in water and soil.	The tiny movers seen in a drop of pond water.
Viruses	Packets of genes with a protein coat; they copy themselves only inside a living host cell.	Baculoviruses used as insect killers (Section: biocontrol).
Viroids	Free RNA without any protein coat; they infect plants.	The agent of potato spindle tuber disease.
Prions	Abnormally folded proteins that force normal proteins to misfold too — no nucleic acid at all.	The agent of "mad cow disease" in cattle.

- **Why the family table matters:** CUET name-pairs almost always name the exact species — so train your eye to attach **which** microbe, not just "a microbe".
- **The borderline members:** viruses, viroids and prions do not grow, eat or divide outside a host — yet they still shape human welfare through disease and biocontrol.

EVERYDAY EXAMPLE — THE INVISIBLE FACTORY

Think of microbes as a giant team of invisible factory workers. You never see them, but give each team its raw material and its digestion becomes your production line: feed them milk and they hand back curd, feed them fruit juice and they hand back alcohol, feed them cow dung and they hand back cooking gas. The mapping is honest — "feed in, digest, product out" is exactly what fermentation is.

WORKED EXAMPLE — WHICH GROUP IS IT?

Classify each worker before moving on — this is the level of recall the chapter builds on:

- **The microbe that sours milk:** a **bacterium** — Lactobacillus, a rod-shaped single cell.
- **The microbe that puffs bread:** a **fungus** — Saccharomyces cerevisiae, the single-celled yeast.
- **The agent of mad cow disease:** a **prion** — a misfolded protein, not even a cell.
- **The potato spindle tuber agent:** a **viroid** — free RNA with no protein coat.

Notice the pattern: three of the four "workers" here are not the microbes we fear. Harm is the exception, not the rule.

Where they live: a quick tour

Habitat	Who lives there	What they are doing
Soil	Billions in every gram of fertile soil	Breaking down dead leaves and recycling nutrients back to plant roots.
Fresh water and seas	Bacteria, algae and protozoa in every drop	Running the base of the food web and cleaning water naturally.
Air	Spores and hardy cells drifting on dust	Travelling — which is how your milk got soured without your help.
Inside living bodies	Gut and skin microbes (including E. coli)	Helping digest food and making vitamins in return for lodgings.
Hot springs	Heat-loving bacteria	Growing happily even near 100 °C — life at the boiling point of water.
Salty marshes	Salt-loving microbes (archaea among them)	Living where ordinary cells would shrivel.

- **The pattern to carry:** wherever there is moisture, food and tolerable conditions, microbes have already moved in — their world is bigger than ours.
- **The welfare lens:** this chapter looks at only a handful of species, but each one has been "employed" by humans — keep asking **who** is working and **what product** comes out.

In the Kitchen — Household Products

Open the fridge and the chapter is already there: curd, cheese, a leftover dosa batter, half a loaf of bread. Every one of these was made by a microbe eating quietly in your own kitchen. These "household products" need no factory — the fermenting vessel is a bowl or a bucket, and the worker is one named microbe. Learn the name-pairs in this section like phone numbers.

- **Curd:** milk + a spoon of old curd (the starter carrying **Lactobacillus**) — the bacteria turn milk sugar into lactic acid, and the acid thickens the milk into curd with its familiar tang.
- **A bonus from curd:** the same Lactobacillus improves the nutritional quality of milk by **increasing its vitamin B_{12}** content.
- **Dosa and idli batter:** ground rice + dal, left overnight — **bacteria and yeasts** ferment the batter together and the CO_2 they release puffs it into a soft, spongy mass.
- **Bread:** the dough is raised by **baker's yeast, Saccharomyces cerevisiae** — it ferments the dough's sugars and releases CO_2, which makes the loaf rise (the little alcohol evaporates in the oven).
- **Toddy:** the milky-looking sap tapped from palm trees, drunk after a few hours of natural **fermentation by yeasts**.

- **Cheese, two famous styles:** Swiss cheese gets its big round holes from the CO_2 made by **Propionibacterium sharmanii**; Roquefort is ripened by growing **fungi** that leave the blue-green veins.

The household name-pairs at a glance

Product	Microbe(s) involved	What actually happens
Curd	**Lactobacillus** (a lactic acid bacterium)	Milk sugar (lactose) → lactic acid; milk thickens and turns tangy; vitamin B_{12} content rises.
Dosa, idli batter	Bacteria + yeasts (a mixed team)	Overnight fermentation releases CO_2 the batter puffs up soft and spongy.
Bread	**Saccharomyces cerevisiae** (baker's yeast)	Sugars in the dough ferment to CO_2 + ethanol; gas bubbles raise the loaf.
Toddy	Yeasts on the palm sap	Sugary palm sap ferments within hours into the milky traditional drink.
Swiss cheese	**Propionibacterium sharmanii**	The bacterium releases CO_2 the trapped gas forms the famous big holes.
Roquefort cheese	Fungi (a Penicillium relative)	Fungal growth ripens the cheese and leaves the blue-green veins.

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- **One trap to flag early:** the holes in Swiss cheese are **bacterial** gas — students love to hand every bubble to yeast. The yeast gets the credit for bread and batter, the bacterium gets the cheese.
- **Same microbe, two jobs:** Saccharomyces cerevisiae appears twice — as **baker's yeast** in the kitchen and as **brewer's yeast** in the beverages industry (next section). One species, two product lines.
- **The starter-culture idea:** a spoon of old curd carries living Lactobacillus into fresh milk — in one line, yesterday's curd hires today's workers. The same logic runs a sewage plant (a pinch of activated sludge reseeds the aeration tank).

WORKED EXAMPLE — SET THE KITCHEN TABLE IN ORDER

A question gives you the product and wants the worker in return. Practise the pull in this direction until it is instant:

- **Curd** → think "lactic acid" → **Lactobacillus** (and remember the vitamin B_{12} bonus).
- **Swiss cheese holes** → think "CO_2 bubbles inside cheese" → **Propionibacterium sharmanii**.
- **Bread dough** → think "risen, smells of fermentation" → **Saccharomyces cerevisiae**.

- **Roquefort veins** → think "blue-green threads" → **fungi ripening the cheese**.

If you can run this list in both directions (product → microbe, microbe → product), the household half of the chapter is yours.

EVERYDAY EXAMPLE — THE KITCHEN IS A MICROBE CAFETERIA

Every household product in this section is one cafeteria, one customer: the microbe eats the raw material on the table, and its "leftover" — acid, gas or flavour — is the product you actually wanted. Milk's sugar becomes lactic acid (curd), dough's sugar becomes carbon dioxide (bread), palm sap's sugar becomes a gently alcoholic drink (toddy). The mapping is direct: what the microbe excretes is what the kitchen serves.

Judge yourself: four true/false calls

- **"Lactobacillus only thickens milk."** — **False.** It also raises the vitamin B_{12} content — a recall line worth keeping.
- **"Bread rises because yeast fills it with ethanol."** — **False.** The rise is CO_2 the tiny amount of alcohol simply evaporates in the oven.
- **"Toddy is fermented palm sap."** — **True.** A few hours of yeast action is the whole recipe.
- **"The veins in Roquefort cheese are bacterial threads."** — **False.** They are **fungal** growth ripening the cheese from inside.

Industrial Products I — Beverages and Antibiotics

A kitchen bowl holds a cup of milk; a factory holds thousands of litres. To make microbes work at that scale, industry grows them in **fermenters** — very large tanks (think 100 to 1000 litres and more) where temperature, air and food are controlled. Inside these steel tanks, the same two household ideas run the show: yeast fermenting sugar, and one microbe's chemistry blocking another's growth.

- **Beverages — the yeast never changes:** whether it is beer, wine or whisky, the worker is **Saccharomyces cerevisiae** (brewer's yeast) — it ferments **malted cereals and fruit juices** into ethanol and CO_2.
- **Why malted cereals:** malting (sprouting and drying grain) converts starch into sugars, and sugar is the only meal yeast can ferment into alcohol.
- **Antibiotics defined:** anti (against) + biotic (life) — an **antibiotic** is a chemical made by one microbe that kills or stops another microbe.
- **The classic pair: Penicillium notatum**, a mould, produces penicillin — the chemical that blocks the growth of **Staphylococcus** bacteria.

The penicillin story — in the right order

Year	Who	What happened
1928	Alexander Fleming	Working with **Staphylococcus** bacteria, he noticed an unwashed culture plate where a mould (**Penicillium notatum**) had landed — and around the mould, the bacteria were dying. He named the chemical penicillin.
1938–1940s	Ernest Chain and Howard Florey	The two scientists isolated penicillin and worked out how to **produce it in quantity** as a medicine.
1942 onwards	Chain, Florey and wartime industry	Penicillin reached patients at scale — its full potential as a drug was established just as World War II made infected wounds a mass problem.
Today	A whole class	Antibiotics (many now made by **other** moulds and bacteria) have brought diseases like **plague, whooping cough, diphtheria and leprosy** under control.

- **The order trap:** Fleming **discovered** (1928); Chain and Florey **developed the drug** (1942). Two different steps, four different names — questions check this pairing again and again.
- **Why it was a revolution:** before antibiotics, a scratch could kill through infection; after them, killer bacterial diseases became treatable within days.
- **Keep the word honest:** an antibiotic works against **microbes** (mostly bacteria). It does nothing to a virus — different enemy, different chemistry.

WORKED EXAMPLE — REBUILD THE STORY FROM A SHUFFLED DECK

The four cards are shuffled: (a) Chain and Florey mass-produce penicillin; (b) Fleming sees mould killing Staphylococcus; (c) antibiotics bring plague, diphtheria and leprosy under control; (d) the mould is named Penicillium notatum and its chemical, penicillin.

- **Correct order:** (b) 1928 observation → (d) the chemical is named → (a) 1942 development into a medicine → (c) the antibiotic era.
- **Check:** Fleming alone only **noticed**; the usable drug arrives only after Chain and Florey; the fall of plague and leprosy is the **consequence**.

Questions test exactly this chain — one event, one year, one name.

Inside the fermenter: what the factory controls

What is controlled	Why it matters to the microbes
Temperature	Each species has its comfort range; too hot or too cold and the production line slows.
pH (acidity)	Acid-building products must not poison the workers — the tank is kept in the friendly zone.
Food (substrate)	Malted cereals, fruit juices or molasses are fed in measured amounts — no sugar, no product.
Air supply	Some jobs need bubbling oxygen, others need none — the tank is run to match the worker.
Sterility	Uninvited microbes eat the raw material and spoil the batch, so tanks are cleaned and sealed.

EVERYDAY EXAMPLE — A BORROWED WEAPON

The antibiotic idea is strange until you see it as borrowed warfare: a mould fights bacteria with a chemical (penicillin), and we simply **borrow that chemical** as our medicine. The mapping holds all the way — the mould is the factory, the chemical is the weapon, and our bloodstream is the battlefield where the bacteria stand no chance.

Industrial Products II — Chemicals, Enzymes and Bioactive Molecules

Fermenters do far more than fill bottles. Fed the right microbe, they pour out food acids, washing-powder enzymes and life-saving molecules. This is the densest name-pair zone of the chapter — four acids, three enzyme jobs, two famous bioactive molecules — so read the tables slowly and let the "why it matters" line stick with each pair.

Organic acids from the fermenter

Acid	Made by	Why it matters
Citric acid	**Aspergillus niger** (a mould)	The sharp taste in fizzy drinks and a common food preservative.
Acetic acid	**Acetobacter aceti** (a bacterium)	The acid of vinegar — cooking and pickling depend on it.
Butyric acid	**Clostridium butylicum** (a bacterium)	An industrial chemical; the same acid gives rancid butter its smell.
Lactic acid	**Lactobacillus** (a bacterium)	The taste of curd — and a workhorse of the food industry.

Enzymes with day jobs

Enzyme	Source microbe	Why it matters
Lipases	Microbial sources (industrial strains)	Mixed into **detergent formulations** to digest oily stains at low wash temperatures.
Pectinases and proteases	Microbial sources (industrial strains)	Added to **clarify bottled juices** — they break the haze-making particles so the juice runs bright.
Streptokinase	**Streptococcus** (a bacterium)	The "clot buster" — given after a heart attack to **remove clots** from blood vessels and reopen flow.

Bioactive molecules — the microbe-made medicines

Molecule	Made by	Why it matters
Cyclosporin A	**Trichoderma polysporum** (a fungus)	An **immunosuppressant** — given to organ-transplant patients so the body's defences calm down and accept the new organ.
Statins	**Monascus purpureus** (a yeast-like mould)	Lower **blood cholesterol** by blocking the enzyme that makes cholesterol in the body.

- **One-line logic for Cyclosporin A:** after a transplant, the immune system treats the new kidney like an invader — Cyclosporin A dials the defences down so the organ is accepted.
- **One-line logic for statins:** cholesterol is **synthesised** inside our body by an enzyme; statins jam that enzyme, so less cholesterol is built — this is why statins are called cholesterol-lowering agents.
- **Do not confuse the two Trichoderma jobs: T. polysporum** gives the medicine (Cyclosporin A), while another **Trichoderma** species works in the field as a biocontrol fungus (later section). Same genus, different jobs.

WORKED EXAMPLE — MATCH THE PAIR UNDER TIME PRESSURE

Cover the middle column and rebuild each pair:

- **Citric acid** → **Aspergillus niger** (mould).
- **Vinegar's acid** → acetic acid → **Acetobacter aceti**.
- **Clot buster** → streptokinase → from **Streptococcus**.
- **Transplant medicine** → Cyclosporin A → **Trichoderma polysporum**.
- **Cholesterol check** → statins → **Monascus purpureus**.

Five pairs, ten facts — this is the exact recall style this topic examines.

If you ran the factory: a quick dispatch table

Order received	Worker to call up
Tonne of citric acid for a soft-drink plant	**Aspergillus niger** in the fermenter
Vinegar batch for the pickle line	**Acetobacter aceti** on ethanol feed
Detergent-grade lipase for a soap company	A lipase-producing microbial strain
Juice-clarifier enzyme for a bottling unit	Pectinases + proteases
Heart-attack kit (clot buster)	Streptokinase from **Streptococcus**
Transplant-patient medicine	Cyclosporin A from **Trichoderma polysporum**
Cholesterol-lowering tablets	Statins from **Monascus purpureus**

- **Notice the shape of the chapter:** the microbe is always the fixed point; only the product changes. Revise by covering the right column and calling the worker from memory.

Sewage Treatment

Every city pipe ends somewhere, and "somewhere" cannot be a river we also drink from. Sewage — the wastewater from homes and industries — is over **99% water**, but the small organic fraction (food bits, human waste, soaps) is food for microbes. Left untreated, that food hits a river, microbes there multiply, and they strip the water of oxygen — fish die first, then the river itself. Treatment plants use the same microbes deliberately, in two stages.

- **BOD — the key number: biochemical oxygen demand** is the amount of oxygen microbes would need to eat all the organic waste in a water sample. **Lower BOD = cleaner water**; a falling BOD is the proof that treatment is working.

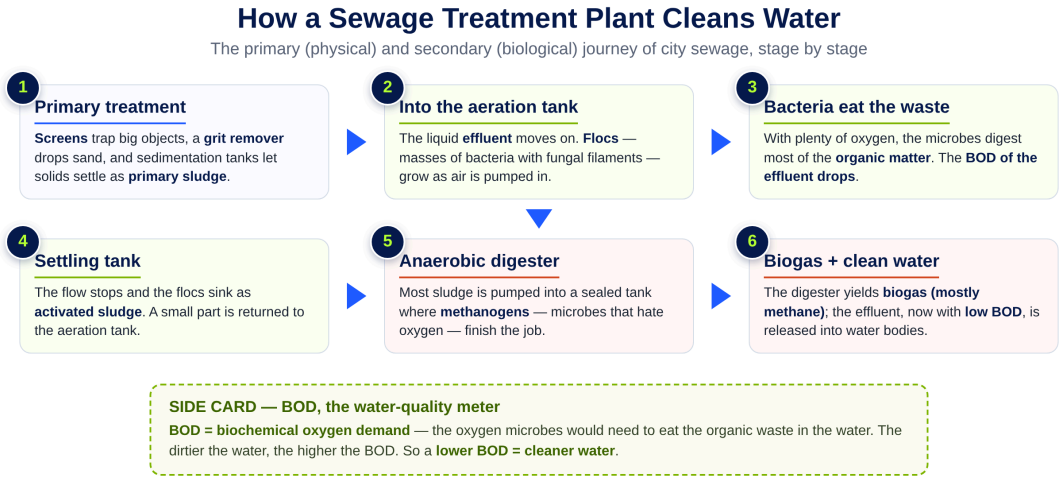


Figure 1: The sewage journey — three physical/biological stops to clean water, two to biogas.

Stage 1 — primary treatment (physical)

- **Screening:** large floating objects (rags, sticks, plastics) are removed by passing sewage through screens.
- **Grit and soil removal:** the flow slows so sand and grit sink first.
- **Sedimentation:** the water then stands in tanks; remaining solids settle to the bottom as **primary sludge**, while the partly cleaned liquid — the **effluent** — flows on.
- **Nothing biological yet:** stage 1 is pure physics — screens, gravity and time.

Stage 2 — secondary treatment (biological)

- **The aeration tank:** the effluent enters a tank where **flocs** — masses of bacteria growing together with fungal filaments — are kept alive with air pumped in from below.
- **Microbes eat the waste:** with oxygen available, the floc bacteria digest most of the organic matter; the **BOD of the effluent falls** sharply.
- **The settling tank:** the water now flows into a quiet tank and the flocs sink, forming **activated sludge** (so named because the settled flocs are still biologically active).
- **Sludge split:** a small part of activated sludge is pumped **back** into the aeration tank as the starter culture; the bulk moves to an **anaerobic sludge digester**.
- **Inside the digester:** anaerobic bacteria — the **methanogens** — digest the sludge and release a gas mixture: **biogas**.
- **Release rule:** the treated effluent is discharged into rivers and streams **only when its BOD is low enough** — the microbes have earned that condition.

The two stages side by side

Stage	Steps in order	Nature	What comes out
Primary	Screening → grit removal → sedimentation	Purely physical — no microbes hired yet	Primary sludge (solids) + effluent (liquid)
Secondary	Aeration tank with flocs → settling tank → anaerobic digester for sludge	Biological — bacteria, fungi, methanogens at work	Activated sludge + biogas + low-BOD effluent

WORKED EXAMPLE — PUT THE SIX STOPS BACK IN ORDER

The cards are shuffled: (a) flocs grow in the aeration tank; (b) grit and sand sink; (c) methanogens digest the sludge; (d) screens catch the rags; (e) flocs sink as activated sludge; (f) BOD falls as bacteria eat the waste.

- **Correct order:** (d) → (b) [primary, physical] → (a) → (f) → (e) [the biological heart] → (c) [sludge finishes underground].
- **Self-check:** screening always opens the sequence; the digester always closes it. If a step mentions air, it belongs to the **secondary** stage.

Any question listing these steps out of order becomes easy once the physical-first, biological-second frame is fixed.

Reading the BOD meter

Water sample	What its BOD is saying
Mountain stream	Almost no organic waste — BOD is very low; microbes there have nothing to eat.
Just below a city drain	Fresh waste has entered — BOD shoots up; microbes multiply and grab every bit of oxygen.
Below the treatment plant	The plant has done its work — BOD is low again, so the water is fit to be released.

- **The river-damage chain in one line:** high-BOD sewage enters a river → river microbes multiply on the feast → dissolved oxygen crashes → fish and other aquatic life are the first to go.
- **Why the 99% figure misleads:** it is the small organic fraction that does the damage — treatment exists to remove that fraction, not the water.

WORKED EXAMPLE — JUDGE THREE STATEMENTS ON BOD

- **"BOD measures the oxygen dissolved in the water."** — **False.** It measures the oxygen microbes **would need** to eat the organic waste — a demand, not a stock.
- **"A falling BOD across the plant shows the treatment is working."** — **True.** Falling demand means the waste food is gone.
- **"Effluent with high BOD can be released safely."** — **False.** High BOD water would trigger the same oxygen crash inside the river.

Biogas

A village cowshed can run a stove. Dung, left to ferment in a sealed tank, gives a clean-burning gas — biogas — and leaves behind a slurry that is still good fertiliser. The workers are the **methanogens**, a group of microbes with one famous quirk: oxygen kills them. That quirk is exactly why the plant is sealed, and why this gas is free for the taking wherever dung, sewage or plant waste piles up.

- **What biogas is:** a mixture — **methane about 50–70%** + carbon dioxide + traces of other gases (such as hydrogen sulphide).
- **Why it is a good fuel:** it burns **without smoke**, leaves no residue, and its heat content is high — methane is the fuel in the mix.
- **Who makes it: methanogens** — anaerobic (oxygen-hating) microbes — acting on dung, sewage and plant waste where air cannot reach.
- **The cattle connection:** a **Methanobacterium** species lives in the **rumen**, the first chamber of a cow's stomach, and helps digest cellulose — so cattle dung arrives pre-loaded with methanogens. That is why gobar (dung) gas is the classic village biogas.

Inside a biogas plant

Part	Job
Concrete tank (built 10–15 feet deep, below ground)	Holds the slurry — dung mixed with water — where fermentation happens without air.
Inlet for slurry	Feeds the fresh dung-water mix into the tank, piped in from the cowshed side.
Floating gas-holder	A drum that rises as gas collects and keeps the tank sealed against air.
Gas outlet	Supplies the biogas straight to the kitchen stove through a pipe.
Spent-slurry outlet	Carries the leftover slurry away — it is still rich in nutrients, so it goes back to the fields as fertiliser.

- **The loop that sells the idea:** dung in → gas out for cooking → slurry out for the field. Nothing in the chain is wasted, which is why biogas plants fit so naturally into village life.

WORKED EXAMPLE — REASON OUT THE COMPOSITION QUESTION

A statement reads: "Biogas is chiefly carbon dioxide with some methane." Judge it.

- **The numbers say otherwise:** methane is the **major** component — about **50–70%** — while carbon dioxide is the second gas and others (like hydrogen sulphide) are only traces.
- **Corrected statement:** biogas is chiefly **methane (50–70%)**, with carbon dioxide and traces of other gases.
- **Bonus check:** the maker microbes are **anaerobic** — any option that puts them "in aerated tanks" is wrong on sight.

Four claims about biogas — judged

Claim	Verdict	The correction
"Biogas is pure methane"	False	Methane is only the major share (50–70%); carbon dioxide and traces ride along.
"Any sealed pit will make gas"	False	The pit needs methanogens in the slurry — no methanogens, no gas.
"Dung is the natural choice of feed"	True	Rumen-dwelling **Methanobacterium** leaves the dung pre-loaded with gas-makers.
"The leftover slurry is a waste problem"	False	The slurry is nutrient-rich fertiliser — the plant's second product.

Why villages adopted the plant

- **Smoke-free kitchens:** biogas burns without the smoke and soot of firewood or dung cakes — cleaner lungs for the household.
- **Waste becomes wealth:** the fuel comes from dung and waste the family already produces — no new purchases, just a tank.
- **The field gains too:** the spent slurry is still nutrient-rich, so it goes back to the land as fertiliser instead of being thrown away.
- **Safe storage by design:** the floating gas-holder keeps the tank sealed — the anaerobic workers stay happy and the gas cannot escape.
- **Two confusions to settle now:** biogas is **not** the same as natural gas (natural gas is a fossil fuel mined from the ground; biogas is made by microbes from fresh waste), and the methanogens of the digester are **not** the floc bacteria of the aeration tank — one team demands zero oxygen, the other demands plenty of it.

Biocontrol Agents

Spray a chemical and you kill the pest **and** the ladybird, the bee and the earthworm. Grow a living enemy of the pest instead, and only the pest falls. That is **biological control** — managing pests with other living things rather than with chemicals — and it keeps the ecology of the field balanced with no toxic residues left behind. Meet the four famous helpers.

- **Bacillus thuringiensis (Bt):** a bacterium sold as dried **spores** — each spore carries a toxin protein that becomes active only inside a caterpillar's gut and kills it. Butterflies and their caterpillars are its targets, and it is essentially harmless to the rest of the field.
- **The gene jump:** the Bt toxin's **genes** were later copied into cotton plants — giving **Bt cotton**, a crop that makes its own caterpillar poison (the details belong to the biotechnology chapter; here, remember the link).
- **Baculoviruses:** insect-killing viruses — the famous one is **nucleopolyhedrovirus** — whose great virtue is that they are **species-specific**: one virus, one pest, nothing else touched.
- **Why baculoviruses suit organic farming:** their narrow attack range leaves birds, beneficial insects, plants and humans untouched — exactly what an organic farmer needs.
- **Trichoderma:** a **free-living fungus** that lives around plant roots and fights several plant **pathogens** — a bodyguard fungus for the roots.
- **The insect helpers: ladybird beetles** eat **aphids** (the sap-sucking pests on stems); **dragonflies** hunt **mosquitoes** over ponds.

The biocontrol roster

Agent	Kind of organism	Target	One-line strength
Bacillus thuringiensis (Bt)	Bacterium	Butterfly caterpillars	Spore-borne toxin that works only in the cater-

Agent	Kind of organism	Target	One-line strength
			pillar gut; its genes later built Bt cotton.
Nucleopolyhedrovirus	Baculovirus (a virus)	One specific insect pest	Species-specific and narrow-spectrum — the organic farmer's insecticide.
Trichoderma	Fungus	Plant root pathogens	Free-living around roots; keeps fungal diseases in check.
Ladybird beetle	Insect	Aphids	Devours sap-sucking aphid colonies on stems.
Dragonfly	Insect	Mosquitoes	Catches mosquitoes on the wing over water.

- **The thinking frame:** whenever a question pairs a biocontrol agent with a target, check the **pairing**, not the word "kills" — every agent here kills something, but each one has exactly one natural target.
- **Chemical vs biological in one line:** chemicals leave residues and kill friend and foe; a biocontrol agent is **living, specific and self-renewing**.
- **Why the Bt toxin needs a gut:** the toxin sits inside the spore as an inactive crystal; only the caterpillar's alkaline gut dissolves it and switches it on — every other stomach simply passes it through.
- **From spore sachet to cotton field:** Bt is sold as spore powder for spraying, **and** its toxin genes live inside Bt cotton — one bacterium, two delivery routes (spray vs plant-built).
- **Keep the two virus names straight: baculovirus** is the family; **nucleopolyhedrovirus** is the member your syllabus names.

WORKED EXAMPLE — MATCH AGENT TO TARGET

Match each agent to its target without looking back:

- **Aphids on a stem** → the **ladybird beetle** (not the dragonfly — it hunts mosquitoes).
- **Caterpillars in a cabbage crop** → **Bt spores** or a suitable **baculovirus**; Trichoplusia (a cabbage looper) is the classic baculovirus host.
- **A root-rot fungus** → **Trichoderma**, the root-bodyguard fungus.
- **Mosquitoes over a pond** → **dragonflies**.

Four matches, four completely different organisms — memorise the pairs as pictures: beetle on the stem, virus in the caterpillar, fungus on the root, dragonfly over water.

Two ways to farm, side by side

Aspect	Chemical pesticide route	Biocontrol route
What is applied	A chemical poison, the same for every target	A living enemy of one particular pest
Whom it kills	Pest plus ladybirds, bees and earthworms in the splash zone	The pest alone — that is the point of specificity
What stays on the crop	Chemical residues that enter the food chain	No toxic residues — the agents are living things from the field itself
Effect on the field's ecology	Population balance is damaged season after season	Ecology stays balanced; pests are held, not erased

- **Organic farming, defined simply:** farming without chemical fertilisers and pesticides — it depends on recycling (compost, manure) and on exactly the biocontrol and biofertiliser teams of this chapter.
- **Where baculoviruses shine:** they are used in **integrated pest management** (IPM) — a programme that mixes methods and reaches for chemicals only as a last resort.
- **One caution worth knowing:** a biocontrol agent is specific, not magic — it controls **its** pest, so fields need a **combination** of agents.

Biofertilisers

A bag of urea feeds the plant today and costs money every season. A **biofertiliser** is a living microbe added to soil or seed that **supplies nutrients** to the plant — often nitrogen — season after season, for free. The supply comes from **nitrogen fixation**, the trick of converting the nitrogen gas of the air into a form roots can absorb. Four teams do this work, and each has its own style.

- **Rhizobium — the symbiotic fixer:** it lives inside **root nodules of legume plants** (peas, beans, gram). The plant feeds it sugars; the microbe hands back fixed atmospheric nitrogen. Both profit — a textbook **symbiosis**.
- **Azotobacter and Azospirillum — the free-living fixers:** these bacteria fix nitrogen **in the soil itself**, without any plant partnership — no nodule, no contract, still useful.
- **Mycorrhiza — the root's fungus: Glomus** wraps around (and into) roots and stretches its threads far into the soil. The plant gains **better phosphorus and water uptake** — and with the water edge comes resistance to drought.
- **Cyanobacteria — the paddy-field team:** blue-green algae such as **Anabaena, Nostoc and Oscillatoria** (plus **Aulosira**, a common fixer in Indian paddy fields) are photosynthetic microbes that **fix nitrogen in flooded rice fields** and also **add organic matter** to the soil.

The biofertiliser table

Type	Organism(s)	Crop / context	Benefit delivered
Symbiotic nitrogen fixer	**Rhizobium**	Legume crops (peas, beans, gram)	Fixes atmospheric nitrogen inside root nodules; both partners gain.
Free-living nitrogen fixer	**Azotobacter**, **Azospirillum**	General soil, any crop	Fixes nitrogen with no plant partner needed.
Mycorrhiza	**Glomus** (a fungus)	Roots of most crop plants	Better phosphorus + water uptake; drought resistance.
Cyanobacteria	**Anabaena**, **Nostoc**, **Oscillatoria**, **Aulosira**	Paddy (flooded rice) fields	Fix nitrogen and add organic matter to the field.

- **Symbiosis vs free-living, in one line:** a nodule on the root means **Rhizobium**; nitrogen fixed in plain soil means **Azotobacter or Azospirillum**.
- **Why phosphorus needs a fungus:** phosphorus barely moves through soil; a fungal web reaching past the root's own zone is the cheapest delivery network a plant can rent.

Four helpers, one comparison glance

Point	Rhizobium	Azotobacter / Azospirillum	Mycorrhiza (Glomus)
Kind of microbe	Bacterium	Bacteria	Fungus
Partnership	Symbiotic — inside legume nodules	Free-living in the soil	Symbiotic — wrapped around roots
Main gift to the plant	Fixed atmospheric nitrogen	Fixed nitrogen, no nodule needed	Phosphorus + water uptake, drought shield

WORKED EXAMPLE — PICK THE RIGHT FIXER

For each field situation, name the helper:

- **A gram field shows healthy root nodules → Rhizobium** (symbiotic fixer in legumes).
- **Plain soil, no nodules anywhere, still nitrogen-rich → Azotobacter / Azospirillum** (free-living).
- **A dry season, yet the crop takes up phosphorus well →** the **mycorrhizal fungus Glomus**.
- **A flooded paddy field turning naturally fertile → cyanobacteria** (Anabaena, Nostoc, Oscillatoria, Aulosira).

The rule: look at the **context** — nodule, plain soil, dry soil, or flooded paddy — and the organism names itself.

EVERYDAY EXAMPLE — THE TENANT FARMER

Rhizobium behaves like a tenant farmer: it moves into the nodule (the house), the legume supplies cooked meals (sugars from photosynthesis), and the tenant pays the rent in fixed nitrogen. Azotobacter, by contrast, is a free worker on the street — no house, no landlord, still handing out nitrogen to whoever's roots are near. The mapping matters because the exam uses the housing detail: nodule = symbiosis, open soil = free-living.

- **Why farmers choose the living helpers:** biofertilisers cost less over the seasons, keep the soil structure healthy, and add no chemical run-off to the nearest pond.
- **How they are used:** cultures are mixed with seeds or worked into the soil before sowing — a step often called **seed treatment** — so the microbes colonise the young roots from day one.
- **Judge four quick statements:** "Azotobacter needs root nodules" — **false**, it is free-living; "mycorrhiza mainly helps phosphorus uptake" — **true**; "cyanobacteria add only nitrogen" — **false**, they also add organic matter; "Rhizobium is free-living" — **false**, it is a nodule symbiont.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — MICROBES IN HUMAN WELFARE

Term / fact	Meaning	Example / number
Curd microbe	Lactobacillus turns milk sugar into lactic acid; also raises vitamin B_{12}	Milk → curd overnight
Swiss cheese holes	CO_2 bubbles released by a bacterium inside the cheese	**Propionibacterium sharmanii**
Baker's yeast	Raises bread dough and ferments batters	**Saccharomyces cerevisiae**
Penicillin discovered	Mould killed Staphylococcus on an unwashed plate	Fleming, 1928; **Penicillium notatum**
Penicillin became a drug	Isolated and mass-produced as medicine	Chain and Florey, 1942
Citric acid	Fermenter acid for drinks and preserves	**Aspergillus niger**
Acetic acid	The acid of vinegar	**Acetobacter aceti**

Term / fact	Meaning	Example / number
Butyric acid	Industrial acid; smell of rancid butter	**Clostridium butylicum**
Lactic acid	The acid of curd	**Lactobacillus**
Lipases	Enzymes mixed into detergent formulations	Digest oily stains
Pectinases + proteases	Clarify bottled juices	Bright, haze-free juice
Streptokinase	Clot buster given after a heart attack	From **Streptococcus**
Cyclosporin A	Immunosuppressant for organ transplants	**Trichoderma polysporum**
Statins	Block the body's cholesterol synthesis	**Monascus purpureus**
BOD	Oxygen microbes would need to eat the organic waste; lower BOD = cleaner water	Effluent released only at low BOD
Primary treatment	Physical: screening, grit removal, sedimentation	Gives primary sludge + effluent
Secondary treatment	Biological: flocs aerated; BOD falls	Gives activated sludge + biogas
Flocs	Masses of bacteria grown with fungal filaments	Sink as activated sludge
Methanogens	Anaerobic microbes that release biogas	Biogas = methane 50–70% + CO_2
Bt	Spore-borne caterpillar toxin; genes went into Bt cotton	**Bacillus thuringiensis**
Baculovirus	Species-specific insect virus insecticide	Nucleopolyhedrovirus; organic farms
Trichoderma (field job)	Fungus guarding roots against pathogens	Free-living in root zones
Rhizobium vs Azotobacter	Nodule symbiont vs free-living soil fixer	Legume nodules vs plain soil
Mycorrhiza	Fungus on roots: better phosphorus + water uptake	**Glomus**; drought shield

Term / fact	Meaning	Example / number
Cyanobacteria	Fix nitrogen in paddy fields; add organic matter	Anabaena, Nostoc, Oscillatoria, Aulosira

Carry these numbers: methane in biogas **50–70%**; sewage is **over 99% water**; penicillin story **1928 → 1942**; BOD rule **lower = cleaner**.

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is actually right	Why / the trap
Blaming the holes in Swiss cheese on yeasts	The holes are CO_2 released by **Propionibacterium sharmanii** — a bacterium	A "yeast for every bubble" habit; cheese holes are bacterial gas.
Saying Fleming developed penicillin into a medicine	Fleming **discovered** it (1928); **Chain and Florey** developed the drug (1942)	Two steps of one story get merged into one name.
Calling flocs "masses of bacteria"	Flocs are masses of bacteria growing **with fungal filaments**	The fungal threads glue the clump — leave them out and the definition is wrong.
Giving methanogens "aerated tanks" to work in	Methanogens are **anaerobic** — free oxygen kills them	Aeration belongs to the floc stage; the digester is sealed.
Saying BOD rises when treatment succeeds	BOD **falls** as microbes eat the organic waste	BOD measures leftover waste food; success means less of it.
Calling streptokinase a "blood thinner"	Streptokinase **dissolves existing clots** after a heart attack	"Clot buster" and "thinner" are different jobs.
Crediting mycorrhiza with nitrogen fixation	Mycorrhiza mainly improves **phosphorus (and water) uptake**	Nitrogen fixing belongs to Rhizobium, Azotobacter and cyanobacteria.

Three checks before you answer:

- **Cyclosporin A dials immunity down** — it is an immunosuppressant, not an infection-fighter; it protects the **transplant**, not the patient from germs.
- **Name the species, not just "a microbe"** — every fact in this topic is a two-part pair (microbe ↔ product); half a pair scores nothing.
- **Sort every sewage step first** — physical (primary) or biological (secondary); the frame alone answers half the questions.

The Whole Chapter on One Page



Figure 2: Chapter mindmap — the six workplaces of microbes and the traps that catch most students.

Practice Questions

Q1. The microbe that turns milk into curd by converting milk sugar into lactic acid is:

(A) Saccharomyces cerevisiae
(B) Lactobacillus
(C) Propionibacterium sharmanii
(D) Acetobacter aceti

Q2. Bread dough rises because baker's yeast ferments the dough's sugars and releases CO_2. The scientific name of baker's yeast is:

(A) Penicillium notatum
(B) Lactobacillus
(C) Saccharomyces cerevisiae
(D) Aspergillus niger

Q3. Citric acid, the sharp taste in fizzy drinks, is produced industrially with the help of:

(A) Acetobacter aceti
(B) Clostridium butylicum
(C) Lactobacillus
(D) Aspergillus niger

Q4. Statins lower blood cholesterol by blocking its synthesis in the body. The microbe from which they are obtained is:

(A) Trichoderma polysporum
(B) Monascus purpureus
(C) Aspergillus niger
(D) Penicillium notatum

Q5. Treated sewage may be released into a river only when its BOD has become low. A low BOD tells you that:

(A) the water carries more organic waste
(B) microbes have raised the oxygen demand
(C) little organic waste is left, so the water is cleaner
(D) the treatment has failed

Q6. During secondary treatment, the masses of bacteria growing together with fungal filaments in the aeration tank are called:

(A) flocs
(B) activated sludge
(C) primary sludge
(D) methanogens

Q7. Biogas is chiefly methane (50–70%). The microbes that produce this gas inside the digester work:

(A) best in bright sunlight
(B) in oxygen-rich slurry
(C) only near 100 °C
(D) in the complete absence of oxygen

Q8. Which free-living fungus, common around plant roots, is used to check plant root pathogens?

(A) Rhizobium
(B) Trichoderma
(C) Glomus
(D) Anabaena

Q9. A legume crop shows healthy root nodules. The nitrogen-fixing symbiont living inside those nodules is:

(A) Rhizobium
(B) Azotobacter
(C) Nostoc
(D) Methanobacterium

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Q10. A patient is given streptokinase shortly after a heart attack. The enzyme works by:

(A) permanently thinning the blood
(B) raising the BOD of the blood
(C) dissolving the clot blocking the vessel
(D) suppressing the immune system

Solutions

Sol. Q1. Curd forms when **Lactobacillus**, a lactic acid bacterium, converts the milk sugar (lactose) into lactic acid — the acid thickens the milk and adds the tang, and also raises vitamin B_{12}. The other pairs belong elsewhere: yeast raises bread, **Propionibacterium** makes the holes in Swiss cheese, and **Acetobacter** gives vinegar its acid.

Answer: (B)

Sol. Q2. Baker's and brewer's yeast are the **same species**: **Saccharomyces cerevisiae**. It ferments sugars to CO_2 + ethanol; the gas raises the loaf. **Penicillium notatum** gives penicillin, **Lactobacillus** sets curd, and **Aspergillus niger** makes citric acid.

Answer: (C)

Sol. Q3. Run the acid table: citric acid ← **Aspergillus niger**; acetic acid (vinegar) ← **Acetobacter aceti**; butyric acid ← **Clostridium butylicum**; lactic acid ← **Lactobacillus**. The fizzy-drink acid is citric, so the maker is the mould **Aspergillus niger**.

Answer: (D)

Sol. Q4. Statins come from **Monascus purpureus** and act by blocking the enzyme that synthesises cholesterol in the body. **Trichoderma polysporum** is the source of Cyclosporin A (the immunosuppressant) — the two fungi are the classic swap-trap here.

Answer: (B)

Sol. Q5. BOD is the oxygen microbes would need to eat the organic waste left in the water. Successful treatment means the microbes have already eaten nearly all the waste, so **little waste remains and the demand for oxygen is low** — hence low BOD = cleaner water.

Answer: (C)

Sol. Q6. In the aeration tank the growing clumps are **flocs** — masses of bacteria associated with fungal filaments. **Activated sludge** is what these flocs become once they sink in the settling tank; primary sludge settles before any biology starts; methanogens live in the anaerobic digester.

Answer: (A)

Sol. Q7. The gas-makers are **methanogens**, anaerobic microbes — free oxygen kills them, which is exactly why the digester is a sealed tank. The mix they release is methane (50–70%) with carbon dioxide and traces of other gases.

Answer: (D)

Sol. Q8. Trichoderma is the free-living fungus of root ecosystems that fights several plant pathogens. **Rhizobium** is a bacterium, **Glomus** is the mycorrhiza fungus (phosphorus and water uptake), and **Anabaena** is a paddy-field cyanobacterium.

Answer: (B)

Sol. Q9. Nodules on legume roots house **Rhizobium** in a symbiosis: the plant supplies sugars, the bacterium supplies fixed atmospheric nitrogen. **Azotobacter** fixes nitrogen without any nodule, **Nostoc** is a cyanobacterium of paddy fields, and **Methanobacterium** lives in the cattle rumen.

Answer: (A)

Sol. Q10. Streptokinase, obtained from **Streptococcus**, is the **clot buster** — it modifies the clot-dissolving system so the blockage in the vessel is digested and blood flow returns. It neither "thins" blood, nor touches BOD, nor suppresses immunity (that is Cyclosporin A's job).

Answer: (C)

FINAL DRILL — SIXTY-SECOND RECALL

Cue	Answer
Curd's microbe and its vitamin bonus	**Lactobacillus** — raises vitamin B_{12} in milk
The gas behind Swiss cheese holes	CO_2 from **Propionibacterium sharmanii**
Yeast of both bread and beer	**Saccharomyces cerevisiae** — one species, two jobs
1928 and 1942 in the penicillin story	Fleming **discovers**; Chain and Florey **develop the drug**
Biogas mix and its workers	Methane 50–70% + CO_2 anaerobic **methanogens**

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